

How to write higher-quality CHI papers (with AI research tools)

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Abstract

Writing high-quality research papers is crucial for advancing your academic career. With the advent of generative artificial intelligence (AI) tools, researchers now have novel ways to improve their writing, literature reviews, and overall paper quality. This course, delivered in person at CHI 2025 in Yokohama, Japan, offers a practical exploration of how to use AI tools effectively throughout the research writing process. Over three interactive 75-minute sessions, participants will learn to apply AI tools to edit their writing, brainstorm ideas, and enhance their paper's readability and impact. Through hands-on activities and peer discussions, attendees will gain the skills needed to produce high-impact CHI papers that meet publication standards. This course emphasizes using AI to support writing, structuring research, and refining contributions, providing attendees with practical tools and insights to succeed in academic publishing in the field of Human-Computer Interaction.

CCS Concepts

• **Applied computing** → *E-learning*; • **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Generative AI, Writing, ChatGPT, Publication, Writing, Submission Process, Research Methods

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1 Benefits

The course's key benefit is helping participants improve their credibility as researchers by producing high-quality papers, enhanced by the effective use of AI tools. In the in-person format, attendees will benefit from direct peer interaction, hands-on practice, and immediate instructor feedback, which will facilitate deeper learning and the application of concepts to their writing. Participants will leave the course with practical skills in using AI to assist their research, as well as increased confidence in structuring and editing their CHI submissions [1, 10, 12, 14]). They will also learn how a

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CHI paper fits into the traditional IMRAD (Introduction, Methods, Research, And Discussion) model, with an in-person focus on interactive activities. Participants will use sticky notes to map out paper structures and discuss carving out a research niche using the CaRS model from Swales [11]. We will also touch upon CHI contribution types as outlined by Wobbrock and Kientz [13], complemented by hands-on demonstrations and peer feedback sessions to reinforce these concepts.

Specifically, I aim for participants to take away the following skills from this course:

- Understanding key AI tools and their functions
- Applying AI tools to generate, organize, and refine ideas
- Analyzing paper structures to identify strengths and areas for improvement
- Evaluating different AI-supported writing techniques for effectiveness
- Creating well-structured research paper sections using AI
- Critically assessing peer feedback and revising drafts
- Using productivity tools for literature review and flowchart creation
- Synthesizing research questions collaboratively during brainstorming activities

This course is an excellent opportunity for CHI attendees to improve their paper-writing skills with generative AI. The plan is for them to gain both practical insights and hands-on experience. Participants will learn specific AI-supported writing techniques and strategies that can make their papers more compelling, clearer, and publication-ready. This in-person experience offers a supportive environment for exploring, discussing, and practicing new skills. The goal is to ensure attendees leave equipped to write impactful CHI submissions with confidence (with an understanding of the ethical boundaries of using generative AI in the writing process) [9].

2 Intended Audience

This in-person course provides an engaging introduction to structuring research papers. It covers how to edit papers, research literature, brainstorm, and draft papers using AI tools—all through interactive, hands-on sessions. No prior experience with AI tools is required for taking the course. So, it is ideal for researchers who want to get better at their writing process with practical AI support. Overall the following people will be a great audience for the course:

- Academics seeking an immersive learning experience with opportunities for real-time collaboration.
- Graduate students and junior researchers who value direct, practical feedback and group exercises to help shape their writing.

- Experienced researchers interested in refining their writing and leveraging AI tools to enhance the clarity and impact of their papers.
- Technical writers, instructors, and other professionals who want to learn about using AI for academic writing.
- Anybody wishing to publish at the CHI conference.

3 Prerequisites

Participants are expected to visit the course website¹ to set up accounts and familiarize themselves with the AI tools prior to attending the sessions. These pre-session instructions will help put everyone on the same page and prepare them to engage with the tools during the in-person, hands-on activities. Participants should also bring a laptop to use during the workshop for exercises involving AI tools.

4 Content

The course is designed to provide a comprehensive guide on how to write better CHI papers through three in-person sessions (see Table 1). These sessions will take participants through all the necessary steps for writing an effective paper with AI support.

Table 1: Structure of three online course units.

Unit 1 CHI Paper Structure	
35 mins	Lecture and Demonstration
20 mins	Small Group Discussion
20 mins	Q&A and Review
Unit 2 Writing Intro and Discussion with AI	
20 mins	AI Research Tools Demonstration
55 mins	Interactive Writing Workshop
Unit 3 Productivity Strategies with AI	
20 mins	Live Demonstrations
25 mins	Sticky Note Brainstorming Activity
30 mins	Hands-On Flowchart Creation

4.1 Unit 1: CHI Paper Structure

4.1.1 Lecture and Demonstration (30 mins). Overview of the CHI paper structure, including the IMRaD format (Introduction [+ Related Work], Methods, Results, and Discussion) and strategies to effectively outline research using the CaRS (Create a Research Space) model. Participants will learn how to establish research territory, identify a research gap, and position their work within the field. This part will guide participants through the CHI paper structure. The instructor will provide examples, outline sections, and show successful approaches to help participants understand how to effectively craft each section of their paper.

4.1.2 Small Group Discussion (20 mins). Break into small groups to discuss paper structures using sticky notes to map out IMRaD and contribution types for an example research topic. Each group will present their findings to the course, allowing for collective feedback and discussion to reinforce understanding.

4.1.3 Q&A and Review (25 mins). Live Q&A to address participant questions, followed by a collaborative group activity where participants refine a provided paper outline using sticky notes and peer suggestions. This activity will help solidify their understanding of effective paper structuring through real-time practice and feedback.

Specifically, we will go over five research questions that every scientific paper must answer:

- **Context:** Why did you do it?
- **Problem:** What did you do?
- **Solution:** How did you do it?
- **Results:** What did you find out?
- **Implications:** What does it all mean?

4.2 Unit 2: Writing Intro and Discussion with AI

4.2.1 Hands-On AI Demonstration (20 mins). Demonstrate key AI tools like Scite AI and Consensus for literature exploration. The instructor will guide participants through practical examples of using these tools to identify relevant literature efficiently. Demo how to draft and edit sections with tools like ChatGPT 4o with canvas and Jenni AI. We will focus on features that enhance clarity and coherence in academic writing.

4.2.2 Interactive Writing Workshop (55 mins). Participants work in pairs to draft parts of an introduction and discussion using AI tools. The instructor will guide real-time exercises where participants create contribution lists, practice editing for conciseness, and refine their drafts. This hands-on session encourages collaboration and immediate application of AI techniques. Here, participants get to practice how to strengthen their writing with constructive peer and instructor feedback.

4.3 Unit 3: Productivity Strategies with AI

4.3.1 Live Demonstrations (20 mins). Demo productivity tools for literature reviews, PDF querying, and brainstorming, including Jenni AI, SciSpace, Litmaps, and Consensus. The instructor will demonstrate how to use these tools to efficiently search for related literature, extract key information from PDFs, and generate relevant ideas. Participants will see real-time examples of how these tools can improve the literature review process and help with productivity in their research.

4.3.2 Sticky Note Brainstorming Activity (25 mins). Each participant will use sticky notes to generate research questions based on their current or planned research projects. We will use a shared AI prompt to improve the research question ideas. These questions will then be shared with their group, who will collaborate to refine and improve them. The goal of this activity is to learn how to frame effective research questions and it supports peer learning by exposing participants to different research topics.

4.3.3 Hands-On Flowchart Creation (30 mins). Use both hand-drawn sketches and AI (such as GPT-4 Vision) to turn flowchart ideas into structured diagrams (with Draw IO). They will practice using AI tools to convert rough sketches into polished digital visual representations. Guided by AI tools, their ability to communicate research processes and results will improve. Working in pairs allows participants to learn from one another and discuss best practices for creating clear and effective flowcharts.

¹<http://chi2025.chicourse.com>

5 Practical Work

The practical work aspect shifts focus this year to active, in-person learning through group activities and exercises. Each unit has been modified to include live demonstrations of AI tools, followed by guided hands-on practice and small group discussions. Participants will use sticky notes to brainstorm and structure content during the sessions. Small group exercises are used to foster discussion and collaboration, encouraging participants to provide feedback on each other's writing. During units two and three, participants practice using AI tools on their own laptops, with guidance from the instructor.

6 Instructor background

Lennart E. Nacke, Ph.D., is a University Research Chair and tenured full Professor for Human-Computer Interaction at the University of Waterloo. He has taught the previous eight iterations of this course at CHI since 2017 Nacke [2, 3, 4, 5, 6, 7, 8] and provides a version of this writing course on his website for purchase: <https://chicourse.com>. He runs the successful *How to Write CHI Papers* podcast and an interview series about writing CHI papers on YouTube. He has served on the SIGCHI program and steering committees and taught University graduate classes on HCI research methods. Dr. Nacke has co-organized many workshops for CHI over the past five years; he also chaired the CHI PLAY 2014 and Gamification 2013 conferences, served as the technical program co-chair for CHI PLAY 2015 and CHI Games and Play subcommittee co-chair for CHI 2017, INTERACT 2019 and CHI PLAY 2022 Full Papers Co-Chair and was the chair of the CHI PLAY steering committee until 2018. He has also reviewed hundreds of papers and had lots of his own submissions rejected from CHI.

7 Resources

Participants can use the following resources to learn about the principles of writing and publishing successful CHI papers:

- How to Write CHI Papers Website
- How to Write CHI Papers Podcast
- How to Write CHI Papers Mailing List
- YouTube: How to Write Papers Channel
- How to Write Better Research Papers Online Course

Printed quick-reference sheets will be provided during sessions to aid hands-on exercises.

8 Accessibility

All materials will be available as recordings and written transcripts so that everyone can take part. This will enable those with hearing impairments to access the content without difficulty. The instructors will also take extra steps to make sure that website materials are

accessible for all users by providing meaningful ALT text for images and using colourblind-safe schemes. The subtitles will make it easy for everyone to follow along with online videos that were made during lessons. These measures taken together help guarantee a more inclusive course experience.

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